Customer Shopping Behavior Analysis Report

Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases from various product categories. The goal is to uncover insights for spending patterns, customer segments, product preferences and subscription behavior to guide a strategic business decision.

Dataset Summary

Rows: 3900
Columns: 18
Key Features:

-- Customer Demographic (Age, Gender, Location).

-- Purchase Details (Item Purchased, Category, Purchase amount, Size, Colour).

-- Shopping Behavior (Discount Applied, Promo code Used, Previous Purchases, Frequency of Purchase, Review Rating, Shipping Type, Subscription Status, Season).

-- Missing Data: 37 values in “Review Rating” column.

EDA using Python

We began with data preparation and cleaning in Python.

Data Loading: Imported dataset using pandas.

Initial Exploration: Used df.head(), df.info() to check structure and df.describe() for summary stats.

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RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object
```

	Customer ID	Age	Gender	Item Purchased	Category	Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Missing Data Handling: Checked for values and found null in Review Rating column and imputed nulls using the median rating of each product category.

Column Standardization: change column cases to snake case for better readability and documentation.

Feature Engineering:

Created age_group column by categorizing customer ages.

Created purchase_frequency_days column from frequency_of_purchases data column.

Data Consistency Check: Checked if discount_applied and promo_code_used were redundant; dropped promo_code_used column.

Database Integration: Downloaded the cleaned DataFrame set into the database for MySQL analysis.

Data Analysis using SQL

Revenue by Gender: Compared total revenue generated by male vs. female.

gender	total_revenue
Male	157890
Female	75191

Top 6 product by average rating: top 6 because 5th and 6th product has the same rating

Item_Name	Average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78
Skirt	3.78

3.78

High-spending discount users: 839 Customers using discount, still spending above average

customer_id	total_revenue
43	100
96	100
194	100
205	100
244	100
249	100

Shipping type comparison: average revenue generated by shipping mode

shipping_type	average_revenue
2-Day Shipping	60.73
Express	60.48
Free Shipping	60.41
Store Pickup	59.89
Next Day Air	58.63
Standard	58.46

Subscribers vs. non-subscribers: Average spent and total spent across subscription status.

subscription_status	count(customer_id)	average_spent	total_spent
Yes	1053	59.49	62645
No	2847	59.87	170436

Discount dependent products: Top 5 product purchased by highest percentage of discount

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer Segmentation: classified customers into new (1 purchase), returning (2-10 purchase) and loyal (>10)

customer_segment	total_customer
Loyal	3116
Returning	701
New	83

Top 3 product by category: Top 3 most purchased product within each category

item_rank	category	item_purchased	total_order
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Relation Between Repeat Buyers and Subscribers: Checked whether customers with >5 purchases more likely to subscribe.

subscription_status	purchase_count
Yes	958
No	2518

Revenue by age group: Revenue contributed by different age group

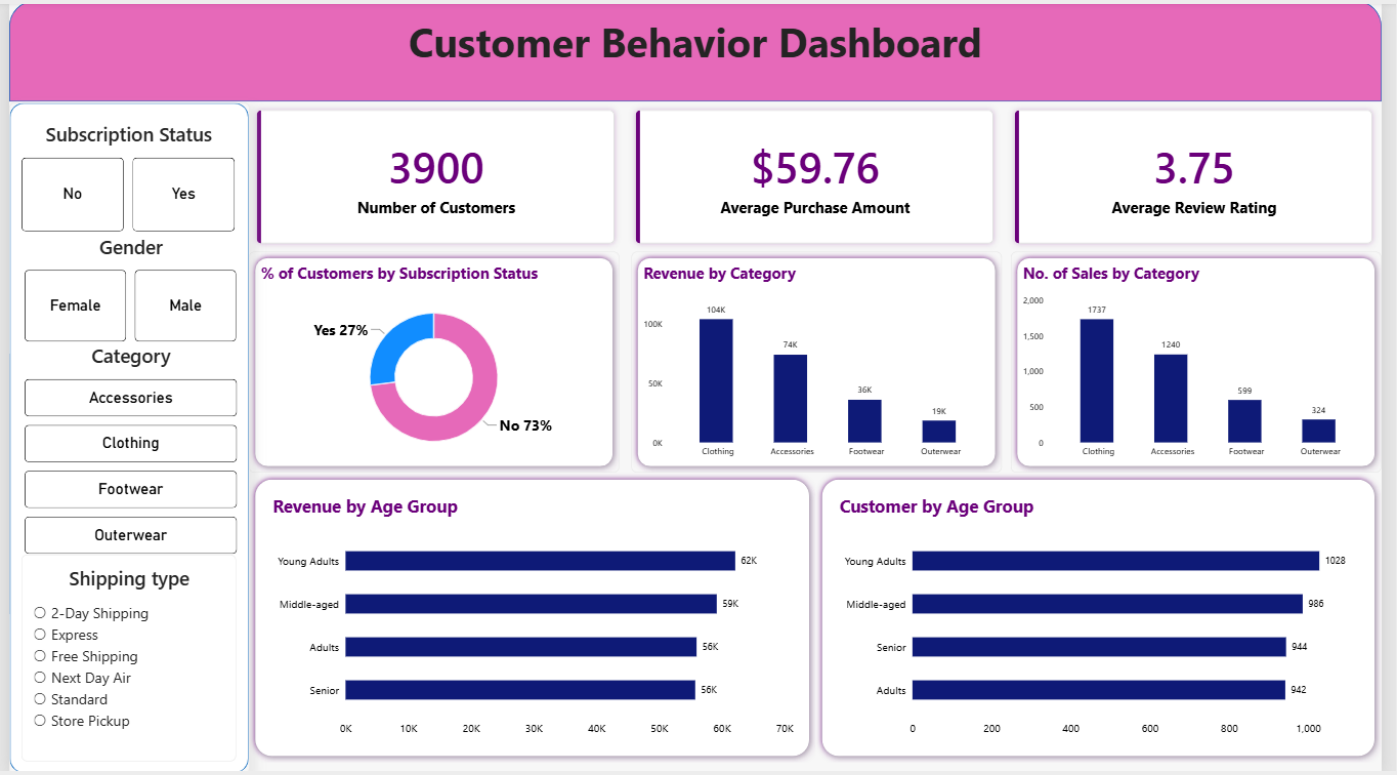
age_group	revenue
Young Adults	62143
Middle-aged	59197
Adults	55978
Senior	55763

Subscription count by Gender: Which gender more likely to purchase a subscription

gender	count(subscription_status)
Male	1053

Dashboard in Power BI

We build an interactive dashboard to present insights visually.



Business Recommendations

- Boost Subscriptions: Promote exclusive benefits for subs to attract more subs (especially female).
- Customer Loyalty Program: Reward returning customers to move them into “Loyal” segment.
- Review Discount Policy: Balance sales boosts with margin control.
- Top Product Promotion: Highlight top-rated and top selling product in campaigns.
- Targeted Marketing: Focus efforts on high revenue contributing age groups and shipping modes.